22.3: Evolution is supported by an overwhelming amount of scientific evidence

In *The Origin of Species*, Darwin marshaled a broad range of evidence to support the concept of descent with modification. Still—as he readily acknowledged—there were instances in which key evidence was lacking. For example, Darwin referred to the origin of flowering plants as an "abominable mystery," and he lamented the lack of fossils showing how earlier groups of organisms gave rise to new groups.

In the last 150 years, new discoveries have filled many of the gaps that Darwin identified. The origin of flowering plants, for example, is much better understood (see Concept 30.3), and many fossils have been discovered that signify the origin of new groups of organisms (see Concept 25.2). In this section, we'll consider four types of data that document the pattern of evolution and illuminate how it occurs: direct observations, homology, the fossil record, and biogeography.

Direct Observations of Evolutionary Change

Biologists have documented evolutionary change in thousands of scientific studies. We'll examine many such studies throughout this unit, but let's look at two examples here.

Natural Selection in Response to Introduced Species

Herbivores, animals that eat plants, often have adaptations that help them feed efficiently on their primary food sources. What happens when herbivores switch to a new food source with different characteristics?

An opportunity to study this question in nature is provided by soapberry bugs, which use their "beak"—a hollow, needlelike mouthpart—to feed on seeds located within the fruits of various plants. In southern Florida, the soapberry bug (*Jadera haematoloma*) feeds on the seeds of a native plant, the balloon vine (*Cardiospermum corindum*). In central Florida, however, where balloon vines have become rare, soapberry bugs feed on the seeds of the golden rain tree (*Koelreuteria elegans*), a species recently introduced from Asia.

Soapberry bugs feed most effectively when the length of their beak is similar to the depth at which seeds are found within the fruit. Golden rain tree fruit consists of three flat lobes, and its seeds are much closer to the fruit surface than are the seeds of the plump, round fruit of the native balloon vine. These differences led researchers to predict that in populations that feed on golden rain tree, natural selection would result in beaks that are *shorter* than those in populations that feed on balloon vine **(Figure 22.13)**. Indeed, beak lengths are shorter in the populations that feed on golden rain tree.

Figure 22.13

Inquiry: Can a change in a population's food source result in evolution by natural selection?

Field Study

Soapberry bugs feed most effectively when the length of their "beak" is similar to the depth of the seeds within the fruit. Researchers measured beak lengths in soapberry bug populations feeding on the native balloon vine. They also measured beak lengths in populations feeding on the introduced golden rain tree. The researchers then compared the measurements with those of museum specimens collected in the two areas before the golden rain tree was introduced.

A soapberry bug with its beak inserted in balloon vine fruit.

Soapberry bug with beak inserted in balloon vine fruit

Results

Beak lengths were shorter in populations feeding on the introduced species than in populations feeding on the native species, in which the seeds are buried more deeply. The average beak length in museum specimens from each population (indicated by red arrows) was similar to beak lengths in populations feeding on native species.

Two graphs show the average beak length in museum specimens from each population.

Two graphs are displayed with x-axis label beak length from 6 to 11 millimeter and y-axis label the number of individuals from 1 to 10. When feeding on native species (balloon vine from southern Florida) a bell shaped curve is shown starting at 7.5 millimeter, peaking at 9.5 millimeter, and trailing off at 11 millimeter (the average was 8.75 millimeters for museum species). When feeding on introduced species (golden rain tree from central Florida) a bell-shaped curve is shown starting at 6 millimeter, peaking at 7 millimeter, and trailing off at 8.5 millimeter (the average was 8.6 millimeter for museum species).

Conclusion

Museum specimens and contemporary data suggest that a change in the seed depth of the soapberry bug's food source can result in evolution by natural selection for a corresponding change in beak size.

Data from S. P. Carroll and C. Boyd, Host race radiation in the soapberry bug: natural history with the history, *Evolution* 46:1052–1069 (1992).

WHAT IF?_Data from additional studies showed that when soapberry bug eggs from a population that fed on balloon vine fruits were then reared on golden rain tree fruits (or vice versa), the beak lengths of the adult insects were most similar to those in the population from which the eggs were initially obtained. Interpret these results._

*For suggested answer, see**Appendix A*.

Researchers have also studied beak length evolution in soapberry bug populations that feed on plants introduced to Louisiana, Oklahoma, and Australia. In each of these locations, the fruit of the introduced plants is larger than the fruit of the native plant. Thus, in populations feeding on introduced species in these regions, researchers predicted that natural selection would result in the evolution of *longer* beaks. Again, data collected in field studies upheld this prediction.

The observed changes in beak lengths had important consequences. In Australia, for example, the increase in beak length nearly doubled the success with which soapberry bugs could eat the seeds of the introduced species. Furthermore, since historical data show that the golden rain tree reached central Florida just 35 years before the scientific studies were initiated, the results demonstrate that natural selection can cause rapid evolution in a wild population.

The Evolution of Drug-Resistant Bacteria

An example of ongoing natural selection that dramatically affects humans is the evolution of drug-resistant pathogens (disease-causing organisms and viruses). This is a particular problem with bacteria and viruses because they can produce new generations in a short period of time; as a result, resistant strains of these pathogens can proliferate very quickly.

Consider the evolution of drug resistance in the bacterium *Staphylococcus aureus*. About one in three people harbor this species on their skin or in their nasal passages with no negative effects. However, certain genetic varieties (strains) of this species, known as methicillin-resistant *S. aureus* (MRSA), are formidable pathogens. Most MRSA infections are caused by recently appearing strains such as clone USA300, which can cause “flesh-eating disease” and potentially fatal infections **(Figure 22.14)**. How did clone USA300 and other strains of MRSA become so dangerous?

Figure 22.14

```
    Clone USA300: a virulent strain of methicillin-resistant
*Staphylococcus aureus* (MRSA).
```

The circular chromosome of clone U S A 300 depicts the complete length of base pairs and adaptive regions.

The circular chromosome of clone U S A 300, containing 2,872,769 base pairs of D N A, and highlighting the regions that increase the strain's virulence. These adaptive regions are as follows:

- Methicillin resistance (between approximately 60,000 to 70,000 base pairs);
- Ability to colonize hosts (between approximately 50,000 to 60,000 base pairs);
- Increased disease severity (between approximately 400,000 to 425,000, 900,000 to 910,000, and 1,900,000 to 1,950,000 base pairs);
- Increased gene exchange (within species) and toxin production (between approximately 1,600,000 to 1,650,000 and 2,100,000 to 2,150,000 base pairs).

Resistant to multiple antibiotics and highly contagious, this strain and its close relatives can cause lethal infections of the skin, lungs, and blood. As shown here, researchers have identified key areas of the USA300 genome that code for adaptations that cause its virulent properties.

WHAT IF? *Some drugs being developed specifically target and kill only* S. aureus*; others slow the growth of MRSA but do not kill it. Based on how natural selection works and on the fact that bacterial species can exchange genes, explain why each of these strategies might be effective.*

*For suggested answer, see**Appendix A*.

The story begins in 1943, when penicillin became the first widely used antibiotic. Since then, penicillin and other antibiotics have saved millions of lives. However, by 1945, more than 20% of the *S. aureus* strains seen in hospitals were already resistant to penicillin. These bacteria had an enzyme, penicillinase, that could destroy penicillin. Researchers responded by developing antibiotics that were not destroyed by penicillinase, but resistance to each new drug occurred rapidly.

For example, a promising new antibiotic, methicillin, was introduced in 1959. But by 1961, methicillin-resistant strains of *S. aureus* were observed. How did these resistant strains emerge? Methicillin works by deactivating an enzyme that bacteria use to synthesize their cell walls. However, some *S. aureus* populations included individuals that were able to synthesize their cell walls using a different enzyme that was not affected by methicillin. These individuals survived the methicillin treatments and reproduced at higher rates than did other individuals. Over time, these resistant individuals became increasingly common, leading to the spread of MRSA.

Initially, MRSA could be controlled by antibiotics that work differently from the way methicillin works. But this has become less effective because some MRSA strains are resistant to multiple antibiotics—probably because bacteria can exchange genes with members of their own and other species. Thus, the multidrug-resistant strains of today may have emerged over time as MRSA strains that were resistant to different antibiotics exchanged genes.

Finally, it is important to note that *S. aureus* is not the only pathogenic bacterium that has evolved resistance to multiple antibiotics. Furthermore, in recent decades, antibiotic resistance has spread much faster than new antibiotics have been discovered—a problem of great public health concern. Hope may loom on the horizon, however. For example, scientists recently reported the discovery of "teixobactin," a new antibiotic that shows promise for treating MRSA and other pathogens. In addition, as we'll describe in the Scientific Skills Exercise in Chapter 27, the methods used in the discovery of teixobactin may lead to the discovery of other new antibiotics as well.

The *S. aureus* and soapberry bug examples highlight three key points about natural selection. First, natural selection is a process of editing, not a creative mechanism. A drug does not *create* resistant pathogens; it *selects for* resistant individuals that are already present in the population. Second, in species that produce new generations in short periods of time, evolution by natural selection can occur rapidly—in just a few years (*S. aureus*) or decades (soapberry bugs). Third, natural selection depends on time and place. It favors those characteristics in a genetically variable population that provide an advantage in the current, local environment. What is beneficial in one situation may be useless or even harmful in another. Beak lengths suitable for the size of the typical fruit eaten by members of a particular soapberry bug population are favored by natural selection. However, a beak length suitable for fruit of one size can be disadvantageous when the bug is feeding on fruit of another size.

BBC Video

A Future Without Antibiotics?

image

Play

image

Play

Homology

A second type of evidence for evolution comes from analyzing similarities among different organisms. As we've discussed, evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered (by natural selection) in its descendants over time as they face different environmental conditions. As a result, related species can have characteristics that have an underlying similarity yet function differently. Similarity resulting from common ancestry is known as **homology**. As we'll describe, an understanding of homology can be used to make testable predictions and explain observations that are otherwise puzzling.

Anatomical and Molecular Homologies

The view of evolution as a remodeling process leads to the prediction that closely related species should share similar features—and they do. Of course, closely related species share the features used to determine their relationship, but they also share many other features. Some of these shared features make little sense except in the context of evolution. For example, the forelimbs of all mammals—including humans, cats, whales, and bats—show the same arrangement of bones from the shoulder to the tips of the digits, even though the appendages have very different functions: lifting, walking, swimming, and flying **(Figure 22.15)**. Such striking anatomical resemblances would be highly unlikely if these structures had arisen anew in each species. Rather, the underlying skeletons of the arms, forelegs, flippers, and wings of different mammals are **homologous structures** that represent variations on a structural theme that was present in their common ancestor.

Figure 22.15

Mammalian forelimbs: homologous structures.

Mammalian forelimbs shown for human, cat, whale and bat.

Even though they have become adapted for different functions, the forelimbs of all mammals are constructed from the same basic skeletal elements: one large bone (purple), attached to two smaller bones (orange and tan), attached to several small bones (gold), attached to several metacarpals (green), attached to approximately five digits, each of which is composed of multiple phalanges (blue).

Mammalian forelimbs shown for human, cat, whale and bat. All vertebrate forelimbs are composed of the same basic bones in the same order, top to bottom: humerus, attached to radius and ulna, followed by carpals, metacarpals and phalanges, but the bones vary by organism reflecting how that animal's limb functions in the environment.

In humans, one large bone (humerus), attached to two smaller bones (radius and ulna), attached to several small bones (carpals), attached to several metacarpals, attached to approximately five digits, each of which is composed of phalanges.

In the cat, the phalanges are shorter and flatter compared to the human forelimb; the humerus is also shorter, and the ulna is longer.

In the whale, the phalanges are much longer than in the human or the cat, filling out the tip of the whale flipper. The humerus, radius, and ulna are all much shorter than they are in the human forelimb.

In the bat, the metacarpals and phalanges are also greatly extended as compared to the human bones, with the phalanges stretching to the edges of the bat wing membrane. The humerus is also shortened.

Comparing early stages of development in different animal species reveals additional anatomical homologies not visible in adult organisms. For example, at some point in their development, all vertebrate embryos have a tail located posterior to (behind) the anus, as well as structures called pharyngeal (throat) arches **(Figure 22.16)**. These homologous arches ultimately develop into structures with very different functions, such as gills in fishes and parts of the ears and throat in humans and other mammals.

Figure 22.16

```
Anatomical similarities in vertebrate embryos.
```

Chick embryo (LM) and human embryo shown with arrows to the pharyngeal arches and post-anal tail.

At some stage in their embryonic development, all vertebrates have a tail located posterior to the anus (referred to as a post-anal tail), as well as pharyngeal (throat) arches. Descent from a common ancestor can explain such similarities.

Some of the most intriguing homologies concern “leftover” structures of marginal, if any, importance to the organism. These **vestigial structures** are remnants of features that served a function in the organism’s ancestors. For instance, snakes arose from ancestors with legs, and the skeletons of some snakes retain vestiges of the pelvis and leg bones of their ancestors. Likewise, blind species of cave fishes descended from ancestors with eyes—which explains why these blind fishes have eye remnants buried under their scales. We would not expect to see these vestigial structures if snakes and blind cave fishes had origins separate from those of other vertebrate animals.

Biologists also observe similarities among organisms at the molecular level. All forms of life use essentially the same genetic code, suggesting that all species descended from common ancestors that used this code. But molecular homologies go beyond a shared code. For example, organisms as dissimilar as humans and bacteria share genes inherited from a very distant common ancestor. Some of these homologous genes have acquired new functions, while others, such as those coding for the ribosomal subunits used in protein synthesis (see Figure 17.18), have retained their original functions. It is also common for organisms to have

genes that have lost their function, even though the homologous genes in related species may be fully functional. Like vestigial structures, it appears that such inactive “pseudogenes” may be present simply because a common ancestor had them.

Homologies and “Tree Thinking”

Some homologous characteristics, such as the genetic code, are shared by all species because they date to the deep ancestral past. In contrast, homologous characteristics that evolved more recently are shared only within smaller groups of organisms.

Consider the *tetrapods* (from the Greek *tetra*, four, and *pod*, foot), the vertebrate group that consists of amphibians, mammals, and reptiles. Like all vertebrates, tetrapods have a backbone. But unlike other vertebrates, tetrapods also have limbs with digits (see Figure 22.15). As suggested by this example, homologous characteristics form a nested pattern: All life shares the deepest layer (in this case, all vertebrates have a backbone), and each successive smaller group adds its own homologies to those it shares with larger groups (in this case, all tetrapods have a backbone *and* limbs with digits). This nested pattern is exactly what we would expect to result from descent with modification from a common ancestor.

Biologists often represent the pattern of descent from common ancestors with an **evolutionary tree**, a diagram that reflects evolutionary relationships among groups of organisms. We will explore evolutionary trees in more detail in Chapter 26, but for now, let’s consider how we can interpret and use such trees.

Figure 22.17 is an evolutionary tree of tetrapods and their closest living relatives, the lungfishes. In this diagram, each branch point represents the most recent common ancestor of the two lineages diverging from that point. For example, lungfishes and all tetrapods descended from ancestor **1**, whereas mammals, lizards and snakes, crocodiles, and birds all descended from ancestor **3**. As expected, the three homologies shown on the tree—limbs with digits, the amnion (a protective embryonic membrane), and feathers—form a nested pattern. Limbs with digits were present in common ancestor **2** and hence are found in all of the descendants of that ancestor (the tetrapods). The amnion was present only in ancestor **3** and hence is shared only by some tetrapods (mammals and reptiles). Feathers were present only in ancestor **6** and hence are found only in birds.

Figure 22.17

```
Tree thinking: information provided in an evolutionary tree.
```

Evolutionary tree for tetrapods with branches explaining evolution of tetrapods, amniotes, reptiles, and birds.

This evolutionary tree for tetrapods and their closest living relatives, the lungfishes, is based on anatomical and DNA sequence data. The purple hatch marks indicate the origin of three important homologies, each of which evolved only once. Birds are nested within and evolved from reptiles; hence, the group of organisms called "reptiles" technically includes birds.

Evolutionary tree for tetrapods. Each branch point represents the common ancestor of the lineages beginning at that point.

Branch point 1: Lungfishes and ancestors with digit-bearing limbs, which includes branch point 2: Amphibians and Amnion ancestors. A subset of Amnion ancestors is branch point 3: mammals and branch point 4: lizards and snakes and branch point 5: crocodiles and ancestors with feathers. Ancestors with feathers include branch point 6: ostriches and hawks and other birds.

Three hatch marks are depicted between branch points 1 and 2, 2 and 3, and 5 and 6, representing homologous characteristics shared by all groups that further branch from the hatch mark.

Tetrapods include amphibians to hawks and other birds. Amniotes include mammals to hawks and other birds. Reptiles include lizards and snakes to hawks and other birds. Birds include ostriches and hawks and other birds.

VISUAL SKILLS *Based on this evolutionary tree, are crocodiles more closely related to lizards or birds? Explain.*

*For suggested answer, see**Appendix A.*

Figure 22.17 Walkthrough

Tree thinking: information provided in an evolutionary tree

Watch an explanation of this figure in the Figure 22.17 Walkthrough: Tree thinking: information provided in an evolutionary tree.

image

Play

image

Play

To explore "tree thinking" further, note that in Figure 22.17 mammals are positioned closer to amphibians than to birds. As a result, you might conclude that mammals are more closely related to amphibians than they are to birds. However, mammals are actually more closely related to birds than to amphibians because mammals and birds share a more recent common ancestor (ancestor **3**) than do mammals and amphibians

(ancestor **2**). Ancestor **2** is also the most recent common ancestor of birds and amphibians, making mammals and birds equally related to amphibians. Finally, note that the tree in Figure 22.17 shows the relative timing of events but not their actual dates. Thus, we can conclude that ancestor **2** lived before ancestor **3**, but we do not know when that was.

Evolutionary trees are hypotheses that summarize our current understanding of patterns of descent. Our confidence in these relationships, as with any hypothesis, depends on the strength of the supporting data. In the case of Figure 22.17, the tree is supported by many different data sets, including both anatomical and DNA sequence data. As a result, biologists are confident that it accurately reflects evolutionary history. Scientists can use such well-supported evolutionary trees to make specific and sometimes surprising predictions about organisms (see Figure 26.17).

A Different Cause of Resemblance: Convergent Evolution

Although organisms that are closely related share characteristics because of common descent, distantly related organisms can resemble one another for a different reason: **convergent evolution**, the independent evolution of similar features in different lineages. Consider marsupial mammals, many of which live in Australia. Marsupials are distinct from another group of mammals—the eutherians, or placental mammals—few of which live in Australia. (Eutherians complete their embryonic development in the uterus, whereas marsupials are born as embryos and complete their development while nursing, often in an external pouch.) Some Australian marsupials have eutherian look-alikes with superficially similar adaptations. For instance, a forest-dwelling Australian marsupial called the sugar glider is superficially very similar to flying squirrels, gliding eutherians that live in North American forests **(Figure 22.18)**. But the sugar glider has many other characteristics that make it a marsupial, much more closely related to kangaroos and other Australian marsupials than to flying squirrels or other eutherians. Once again, our understanding of evolution can explain these observations. Although they evolved independently from different ancestors, these two mammals have adapted to similar environments in similar ways. In such examples in which species share features because of convergent evolution, the resemblance is said to be **analogous**, not homologous. Analogous features share similar function, but not common ancestry, while homologous features share common ancestry, but not necessarily similar function.

Figure 22.18

```
Convergent evolution.
```

Sugar glider from Australia and flying squirrel from North America. They look similar.

The ability to glide through the air evolved independently in these two distantly related mammals.

The Fossil Record

A third type of evidence for evolution comes from fossils. The fossil record documents the pattern of evolution, showing that past organisms differed from present-day organisms and that many species have become extinct. Fossils also show the evolutionary changes that have occurred in various groups of organisms. To give one of hundreds of possible examples, researchers found that over several thousand years, the pelvic bone in fossil stickleback fish became greatly reduced in size. The consistent nature of this change over time suggests that the reduction in the size of the pelvic bone may have been driven by natural selection.

Fossils can also shed light on the origins of new groups of organisms. An example is the fossil record of cetaceans, the mammalian order that includes whales, dolphins, and porpoises. Some of these fossils **(Figure 22.19)** provided strong support for a hypothesis based on DNA sequence data: that cetaceans are closely related to even-toed ungulates, a group that includes hippopotamuses, pigs, deer, and cows.

Figure 22.19

```
Ankle bones: one piece of the puzzle.
```

Most mammals represented by a Canis (dog); Cetaceans and even-toed ungulates represented by (b) Pakicetus, (c) Sus (pig), (d) Odocoileus (deer).

*Comparing fossils and present-day examples of the astragalus (a type of ankle bone) indicates that cetaceans are closely related to even-toed ungulates. **(a)** In most mammals, the astragalus is shaped like that of a dog, with a double hump on one end (red arrows) but not at the opposite end (blue arrow). **(b)** Fossils show that the early cetacean *Pakicetus* had an astragalus with double humps at both ends, a shape otherwise found only in pigs **(c)**, deer **(d)**, and all other even-toed ungulates.*

What else can fossils tell us about cetacean origins? The earliest cetaceans lived 50–60 million years ago. The fossil record indicates that prior to that time, most mammals were terrestrial. Although scientists had long realized that whales and other cetaceans originated from land mammals, few fossils had been found that revealed how cetacean limb structure had changed over time, leading eventually to the development of flippers as well as to the loss of hind limbs and the development of flukes (the lobes on a whale's tail). In the past few decades, however, a series of remarkable fossils have been discovered in Pakistan, Egypt, and North America. These fossils document steps in the transition from life on land to life in the sea, filling in some of the gaps between ancestral and living cetaceans **(Figure 22.20)**.

Figure 22.20

```
The transition to life in the sea.
```

An evolutionary tree.

Evolutionary tree with branch point 1: other even-toed ungulates and branch point 2 (56 million years ago); branch point 2: hippopotamuses and common ancestors of cetaceans (54 million years ago). Common ancestors of cetaceans include branch point 3 (52 million years ago): Pakicetus (extinct 45 million years ago) and branch point 4 (48 million years ago). Branch point 4 includes Rodhectus (extinct 40 million years ago) and branch point 5 (40 million years ago) which includes Dorudon (extinct 35 million years ago) and living cetaceans. All cetaceans have a pelvis. Hippopotamuses, Pakicetus, Rodhocetus, and Durodon have also have a femur, tibia, and foot.

Multiple lines of evidence support the hypothesis that cetaceans (yellow labels) evolved from terrestrial mammals. Fossils document the reduction over time in the pelvis and hind limb bones of extinct (†) cetaceans, including *Pakicetus*, *Rodhocetus*, and *Dorudon*. DNA sequence data support the hypothesis that cetaceans are most closely related to hippopotamuses. (Note that the drawings are not to scale.)

VISUAL SKILLS *Use the diagram to determine which happened first during the evolution of cetaceans: changes in hind limb structure or the origin of tail flukes (the lobes on a whale's tail). Explain.*

*For suggested answer, see*Appendix A*.*

Figure 22.20 Walkthrough

The transition to life in the sea

Watch an explanation of this figure in the Figure 22.20 Walkthrough: The transition to life in the sea.

image

Play

image

Play

Finally, although living cetaceans look very different from living even-toed ungulates, the fossil discoveries show that extinct early cetaceans such as *Pakicetus* looked fairly similar to deer and other living even-toed ungulates. The extinct early cetaceans also looked very similar to extinct early even-toed ungulates, such as *Diacodexis***(Figure 22.19)**. Similar patterns are seen in fossils documenting the origins of other groups of organisms, including mammals (see Figure 25.7), flowering plants (see Concept 30.3), and tetrapods (see Figure 34.21). In each of these cases, the fossil record shows that over time, descent with modification

produced increasingly large differences among related groups of organisms, ultimately resulting in the diversity of life today.

Figure 22.21

```
*Diacodexis*, an early even-toed ungulate.
```

Diacodexis, an early even-toed ungulate (length including tale is approximately 80 centimeters; height is approximately 30 centimeters).

Biogeography

A fourth type of evidence comes from **biogeography**, the scientific study of the geographic distributions of species. These distributions are influenced by many factors, including *continental drift*, the slow movement of Earth's continents over time. About 250 million years ago, these movements united all of Earth's landmasses into a single large continent called **Pangaea** (see Figure 25.16). Roughly 200 million years ago, Pangaea began to break apart; by 20 million years ago, the continents were within a few hundred kilometers of their present locations.

We can use our understanding of evolution and continental drift to predict where fossils of different groups of organisms might be found. For example, scientists have constructed evolutionary trees for horses based on anatomical data. These trees and the ages of fossils of horse ancestors suggest that the genus that includes present-day horses (*Equus*) originated 5 million years ago in North America. Geologic evidence indicates that at that time, North and South America were not yet connected, making it difficult for horses to travel between them. Thus, we would predict that the oldest *Equus* fossils should be found only on the continent on which the group originated—North America. This prediction and others like it for different groups of organisms have been upheld, providing more evidence for evolution.

Evolution also explains biogeographic data. For example, a group of three species of freshwater fish in the family Galaxiidae are found in regions separated by wide stretches of open ocean **(Figure 22.22)**. None of these three species can survive in salt water, so how can we explain their geographic distributions? These three species represent all known living descendants of a common ancestor. Since all of the living species cannot survive in salt water, it is likely that their common ancestor also was restricted to freshwater habitats. At first glance, this only deepens the mystery—how could that ancestor or its descendants have swum across vast stretches of ocean? A clue comes from genetic analyses indicating that the evolutionary lineages leading to these three species began to diverge about 55 million years ago. By that time, the southernmost portion of Pangaea was breaking apart into what would eventually become South America, Australia, and Antarctica.

This suggests that their ancestors had spread across southern Pangaea before it broke apart. At later times, these three freshwater species originated on the continental regions where they currently live, resulting in their present-day geographic distribution.

Figure 22.22

Closely related freshwater fish separated by 9,000 km of ocean.

On a tree, Aplochiton (two species; South America) and Lovettia (one species; Tasmania, Australia) share a common ancestor about 55 million years ago. They all share a common ancestor with other genera in the Galaxiidae approximately 69 million years ago.

There are three species in these two closely related genera of freshwater fish within the family Galaxiidae.

What Is Theoretical About Darwin's View of Life?

Some people dismiss Darwin's ideas as "just a theory." But the *pattern* of evolution—the observation that life has evolved over time—is supported by a great deal of evidence. In addition, Darwin's explanation of the *process* of evolution—that natural selection is the primary cause of the observed pattern of evolutionary change—makes sense of massive amounts of data. As the **Scientific Skills Exercise** describes, the effects of natural selection also can be observed and tested in nature.

Scientific Skills Exercise

Making and Testing Predictions

AP Science Practices 2, 3, 4, 6

Can Predation Result in Natural Selection for Color Patterns in Guppies? Our understanding of evolution changes constantly as new observations lead to new hypotheses—and hence to new ways to test our understanding of evolutionary theory. Consider the wild guppies (*Poecilia reticulata*) that live in pools connected by streams on the Caribbean island of Trinidad. Male guppies have highly varied color patterns that are controlled by genes that are only expressed in adult males. Female guppies choose males with bright color patterns as mates more often than they choose males with drab coloring. But the bright colors that attract females also can make the males more conspicuous to predators. Researchers observed that in pools with few predator species, the benefits of bright colors appear to "win out," and males are more brightly colored than in pools where predation is more intense.

One guppy predator, the killifish, preys on juvenile guppies that have not yet displayed their adult coloration. Researchers predicted that if adult guppies with drab colors were transferred to a pool with only killifish, eventually the descendants of these guppies would be more brightly colored (because of the female preference for brightly colored males).

Guppies are transplanted from pools with pike-cichlids and guppies to pools with killifish, but no guppies prior to the transplant.

How the Experiment Was Done Researchers transplanted 200 adult guppies from pools containing pike-cichlid fish, predators of adult guppies, to pools with only killifish, predators of juvenile guppies. They tracked the number of bright-colored spots and the total area of those spots on adult male guppies in each generation.

Data from the Experiment After 22 months (15 generations), researchers compared the color pattern data for guppies from the source and transplanted populations.

Two bar charts show (1) number of colored spots and (2) area of colored spots (millimeters squared) on the y-axis and the source population and transplanted population on the x-axis.

Two bar charts show (1) number of colored spots and (2) area of colored spots (millimeters squared) on the y-axis and the source population and transplanted population on the x-axis.

- Chart 1: Source population has 8 colored spots and the transplanted population has 12.
- Chart 2: Source population has an area of colored spots of 5 millimeters squared and the transplanted population has 11.

Data from J. A. Endler, Natural selection on color patterns in *Poecilia reticulata, Evolution* 34:76–91 (1980).

INTERPRET THE DATA

- Identify the following elements of hypothesis-based science in this example:
- question,
- hypothesis,
- prediction,
- control group, and
- experimental group. (For additional information about hypothesis-based science, see Concept 1.3 and the Scientific Skills Review in Appendix D.)

- Explain how the types of data the researchers chose to collect enabled them to test their prediction.
- What conclusion do you draw from the data presented above?
-
    - Predict what would happen if, after 22 months, guppies from the transplanted population were returned to the source pool.
- Describe an experiment to test your prediction.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

What, then, is theoretical about evolution? Keep in mind that the scientific meaning of the term *theory* is very different from its meaning in everyday use. The colloquial use of the word *theory* comes close to what scientists mean by a hypothesis. In science, a theory is much more comprehensive than a hypothesis. A theory, such as the theory of evolution by natural selection, accounts for many observations and explains and integrates a great variety of phenomena. Such a unifying theory does not become widely accepted unless its predictions stand up to thorough and continual testing by experiment and additional observation (see Concept 1.3). As the rest of this unit demonstrates, this has certainly been the case with the theory of evolution by natural selection.

The skepticism of scientists as they continue to test theories prevents these ideas from becoming dogma. For example, although Darwin thought that evolution was a very slow process, we now know that this isn't always true. Populations can evolve rapidly, and new species can form in relatively short periods of time: a few thousand years or less. Furthermore, evolutionary biologists now recognize that natural selection is not the only mechanism responsible for evolution. Indeed, the study of evolution today is livelier than ever as scientists use a wide range of experimental approaches and genetic analyses to test predictions based on natural selection and other evolutionary mechanisms.

Although Darwin's theory attributes life's diversity to natural processes, the diverse products of evolution are nevertheless elegant and inspiring. As Darwin wrote in the final sentence of *The Origin of Species*, "There is grandeur in this view of life… [in which] endless forms most beautiful and most wonderful have been, and are being, evolved."